

Deserialization of untrusted data — CodeQL query help documentation

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- Missing a summary in documentation comment
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Deserialization of untrusted data

ID: cs/unsafe-deserialization-untrusted-input

Kind: path-problem

Security severity: 9.8

Severity: error

Precision: high

Tags:

- security
- external/cwe/cwe-502

Query suites:

- csharp-code-scanning.qls
- csharp-security-extended.qls
- csharp-security-and-quality.qls

[Click to see the query in the CodeQL repository](#)

Deserializing an object from untrusted input may result in security problems, such as denial of service or remote code execution.

Note that a deserialization method is only dangerous if it can instantiate arbitrary classes. Serialization frameworks that use a schema to instantiate only expected, predefined types are generally not tracked by this query. Such frameworks are generally safe with respect to arbitrary-class-instantiation and gadget-chain attacks when the schema is trusted and does not permit user-controlled type resolution. However, care must be taken to ensure the schema strictly limits the allowed types. Permitting common standard library classes can still leave the application vulnerable to gadget-chain attacks.

Example

In this example, text from an HTML text box is deserialized using a `JavaScriptSerializer` with a simple type resolver. Using a type resolver means that arbitrary code may be executed.

```
using System.Web.UI.WebControls;
using System.Web.Script.Serialization;

class Bad
{
    public static object Deserialize(TextBox textBox)
    {
        JavaScriptSerializer sr = new JavaScriptSerializer(new SimpleTypeResolver());
        // BAD
        return sr.DeserializeObject(textBox.Text);
    }
}
```

To fix this specific vulnerability, we avoid using a type resolver. In other cases, it may be necessary to use a different deserialization framework.

```
using System.Web.UI.WebControls;
using System.Web.Script.Serialization;

class Good
{
    public static object Deserialize(TextBox textBox)
    {
        JavaScriptSerializer sr = new JavaScriptSerializer();
        // GOOD: no unsafe type resolver
        return sr.DeserializeObject(textBox.Text);
    }
}
```

In the following example potentially untrusted stream and type is deserialized using a `DataContractJsonSerializer` which is known to be vulnerable with user supplied types.

```
using System.Runtime.Serialization.Json;
using System.IO;
using System;

class BadDataContractJsonSerializer
{
    public static object Deserialize(string type, Stream s)
    {
        // BAD: stream and type are potentially untrusted
        var ds = new DataContractJsonSerializer(Type.GetType(type));
        return ds.ReadObject(s);
    }
}
```

To fix this specific vulnerability, we are using hardcoded Plain Old CLR Object ([POCO](#)) type. In other cases, it may be necessary to use a different deserialization framework.

```

using System.Runtime.Serialization.Json;
using System.IO;
using System;

class Poco
{
    public int Count;

    public string Comment;
}

class GoodDataContractJsonSerializer
{
    public static Poco Deserialize(Stream s)
    {
        // GOOD: while stream is potentially untrusted, the instantiated type is
        hardcoded
        var ds = new DataContractJsonSerializer(typeof(Poco));
        return (Poco)ds.ReadObject(s);
    }
}

```

References

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Muñoz, Alvaro and Mirosh, Oleksandr: [JSON Attacks](#).

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Common Weakness Enumeration: [CWE-502](#).

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